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Transitioning to an AI-enabled Generative Cyber-Physical Servicescape

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Transitioning to an AI-enabled Generative Cyber-Physical Servicescape

Abstract

Customers of various services have ever-changing expectations from service providers, which is placing critical pressure on providers to develop a synchronous, quick system that positively impacts user experiences while ensuring that providers are efficient, like ridesharing, home care, vehicle repairing, etc. To serve customers seamlessly and offer enhanced experience, businesses are turning to generative AI. Hence, this study focuses on conceptualizing an AI-enabled generative Cyber-Physical Servicescape (CPSC) structure and how different elements are related. This study adopted a grounded theory approach and used semi-structured interviews to collect data from 30 industry experts. First-order concepts, second-order themes, and aggregate dimensions are developed based on the interview data. This analysis further led to the emergence of four core elements of an AI-enabled generative CPSC: user interface, cybercore, servicescape design, and AI-based CPS servicescape. This study contributes to theory by developing four propositions and a framework that indicates the relationship between the four emerging elements of an AI-enabled generative CPSC. This study offers implications for theory and practice that can be considered when developing cyber-physical service systems in services and employing generative AI for best customer service.

Keywords: Service Systems; Cyber-Physical Systems; Cyber-Physical Servicescape; Generative Artificial Intelligence; Digital Transformation

1. Introduction

Apart from goods, services compose a significant part of world trade and economy. Services are distinct from production processes in that they are intangible, experiential, and co-produced (Lee et al., 2023; Margherita & Braccini, 2020; Soto Setzke et al., 2023). Services are process-driven and require various actors and systems to work in an alliance for unparalleled experience for consumers (Lee et al., 2023; Lim et al., 2018; Lempinen & Rajala, 2014; Setia et al., 2013). For instance, UBER offers its customers fast, flawless, and efficient transportation within a particular city by involving riders and drivers. Unlike production systems, where a customer's evaluation happens after a product is consumed, a service experience is simultaneously created for consumption at the delivery point (Karwatzki et al., 2022; Guan et al., 2021). For instance, food in a restaurant may be delicious, but it may create a bad impression if customers have to

wait too long. A user and a service provider co-create a service experience, and users' satisfaction depends on how the service is delivered (Winkler & Wulf, 2019; Xie et al., 2016). For example, in a hotel stay, how a traveler is treated by a travel agency, hotel and other stakeholders involved defines the customer's experience.

Traditionally, services had a human element as the major interface between the creation and delivery of services (Knote et al., 2021). Co-generation involving human beings with technology is inevitable in services such as banking, mobile payments, insurance, hospitality, healthcare, security, and so on (Kar, 2021; Sahaym et al., 2022). Customer engagement and experience are of key importance while developing service systems (Xie et al., 2016). Service systems are defined as "a configuration of people, technologies, and other resources that interact with each other to create mutual value" (Maglio et al., 2009, p.395). Hence, before offering a service, the role of technology and how it is going to interact with users should be clear (Lee et al., 2023; Skaggs & Youndt, 2004). For example, customers may interact with the services offered through social media, websites, phone apps, and conversations with a support team to resolve and address their concerns or expectations from a service (Forsgren & Byström, 2018; Setia et al., 2013). Firms have also started embracing Generative Artificial Intelligence for automating these interactions (Modgil et al., 2025). Emerging technologies such as Artificial Intelligence (AI) based applications like chatbots are further shaping these interactions to improve overall customer experience (Kushwaha et al., 2021). In services, the servicescape concept (Bitner, 1992) has been widely used in the extant literature and practice. A *servicescape* is defined as the physical environment in which a service is offered. It includes all the physical features, such as lighting, colors, symbols, artifacts, and overall layout, in the arrangement of various tools and equipment (Zomerdijk & Voss, 2010).

Since servicescapes involve humans conceptualizing service experiences, much emphasis in the literature has been placed on imparting interpersonal (Knote et al., 2021), task-based (Setia et al., 2013), and empathy skills (Guan et al., 2021) to service managers. For instance, interpersonal skills help service managers listen, respond, and effectively address the needs of the customer. In contrast task-based skills indicate task prioritization, schedule, and delegate the work, and empathy skills are a way of validating customers' point of view (Knote et al., 2021). However, with the advancement of technologies, the human element is gradually being replaced by machines, which is changing the servicescape concept. For instance, automated kiosks, teller machines, e-service kiosks, appointment systems, and robotic surgeries have

become common and can be referred as non-AI-based generative systems. In the recent past, a shift is observed toward AI-based technologies such as ChatGPT[1] (Dowling & Lucey, 2023), which paves the way for intelligent systems that can think and perform like humans (Rezwana & Maher, 2022). AI-based-generative service systems employ technology to create content and solutions that autonomously executed responding to a particular scenario. AI-based-generative systems can help optimize the processes, and personalized interactions, as compared to non-AI-based generative service systems, which employs predefined rules and heuristics to respond to certain situations and scenario (Dwivedi et al., 2024). Some of the studies have flagged the hype of the generative AI while conducting field experiments (Dell'Acqua et al., 2023), where they found the implications of AI for content generation and quick response, rather than solving any technical and complex problems. The non-AI-based generative systems cannot train themselves and learn from the data. In nutshell the degree of adaptability, complexity of solutions, data dependency, flexibility and scalability is higher in AI-based generative systems than non-AI-based generative systems.

The rising digitization has given birth to cyber-physical systems (CPS) (Huber et al., 2022), where it involves a holistic integration of physical and digital systems besides the complex processes that bind the two seamlessly (Nguyen & Nof, 2019). CPS are sociotechnical; they offer a range of computational abilities that seamlessly integrate physical and digital realms for digital transformation (Soto Setzke et al., 2023). Such systems employ contextual awareness to incorporate dynamic changes in an environment and adjust service system responses to suit business and user needs (Muthukannan et al., 2021).

Earlier research has highlighted that CPS solve challenges in integrating physical and digital systems and their processes in varied contexts and user requirements (Karwatzki et al., 2022). Furthermore, Tuunanen et al. (2019) argued that creating a CPS-enabled service system requires a detailed analysis of interactions and processes and considering the roles and skills of the various actors/users involved. The organization of key elements and the presence of key elements make a service system useful for users (Huber et al., 2019; Li & Tuunanen, 2022). With the emergence of generative AI systems, the services landscape is getting a new shape (Dwivedi et al., 2024; Ooi et al., 2023), witnessing the impact in different industries. However, the literature is scant in highlighting the role of AI in general and Generative AI in particular

[1] ChatGPT is an artificial intelligence chatbot, https://en.wikipedia.org/wiki/ChatGPT.

in developing CPS servicescape and its organization (Choudhary et al., 2024). Motivated by this gap, this research explored the following research questions:

1. *What are the key elements of an AI-enabled Generative Cyber-Physical Servicescape*
2. *How should Generative Cyber-Physical Servicescape be organized?*

To answer these research questions, we interviewed professionals. Consequently, grounded theory methodology (Gioia et al., 2013) was the way forward, given that this ecosystem is nascent regarding documented theoretical evidence. Data were collected from 30 respondents working in the space of CPS and services in different industries (See Appendix A) and processed rigorously to identify first-order concepts, second-order themes, and aggregate dimensions (Gioia et al., 2013), which further led to the development of an AI-enabled generative CPSC. This paper is organized as follows: Section 2 presents the theoretical background. The research design is covered in section 3. The findings are presented in section 4. Further, the discussion is presented in section 5.

2. Theoretical Background

This section highlights the role of systems in developing services and how does contextual consideration, people orientation, and continual improvements through iterative design can influence services. Further, the role of role of cyber-physical systems and generative AI is highlighted in cyber-physical systems and how do that lead towards AI-enabled generative CPSC.

2.1. Service Systems

With the growth of the Internet, most services have shifted from being physically served to being an online mode and hybrid. The designs and environments of the physical mode of services play a critical role, and the same applies to online services (Chandra et al., 2012; Kohler et al., 2011). However, modern service systems integrate an organization's technology, architecture, and stakeholders to fulfill end consumers' aspirations and co-create value (Maglio et al., 2009; Margherita & Braccini, 2020; Peters et al., 2015). Service systems connect internal and external information to co-create configurations of people, views, data, and technology (Chuang & Lin, 2017; Setia et al., 2011; Xu et al., 2013), leading towards digital transformation. This co-creation depends on processes, flows, and designs (Li & Tuunanen, 2022). An online environment consists of architecture and flow of typical transactions

generating tremendous data requiring analytics in services (Versteeg & Bouwman, 2006). Hence, designing a service system that ensures high-quality interactions between stakeholders is critical to co-create value (Tung et al., 2014). For instance, ride-sharing system like Uber has different service providers, including drivers, technology providers etc., to co-create value. In this system, customers are passengers, and service systems enable matching the customer to the right driver and facilitate payment and feedback mechanism, where it leverages both virtual and physical environments and uses resources such as vehicles, the Uber app, and GPS technology.

Service systems have been proposed as a priority in research around services (Li & Tuunanen, 2022; Seidmann et al., 2011; Tung et al., 2014; Zomerdijk & Voss, 2010), with a key emphasis on leveraging technology to design better systems. A significant part of service systems involves the design of a servicescape (Bitner, 1992); in the case of online services, it is the design of the system, website, and application (Wulf & Blohm, 2020). System concept and design are instrumental in providing cues to consumers, influencing their affective and cognitive states and subsequent service experiences (Bitner, 1992; Tung et al., 2014; Winkler & Wulf, 2019).

On a broader level, the service systems should be based on insights about the customers' requirements, context, and culture. These, in turn, are used to transform and formulate key interactions in a service system digitally (Seidmann et al., 2011), and customer input should be sought to improve the design (Boudreau & Lakhani, 2013; Setia et al., 2013). Traditional service systems are centralized and are driven by static and predefined algorithms, with limited scope for adaptability and personalization. Such service systems are capable of service in repetitive task environments. However, in many services today, dynamism and complexity cannot be ignored (Dwivedi et al., 2024). This inadequacy of traditional service systems creates a base for the development of AI-enabled generative systems that can integrate physical and digital environments, quickly adapt to different scenarios and have enough flexibility and scalability to handle load (Ferraro et al., 2024; Ooi et al., 2023; Richey Jr et al., 2023). Incorporating such considerations can help create innovative systems that offer customers a unique experience and take advantage of intelligent technologies to build independent and efficient service delivery mechanisms. Service systems, in general, accounts for the following considerations:

2.1.1. Contextual consideration

The overall environment offered to customers and stakeholder define the service experience. The interaction between a customer and a system defines customer retention (Zomerdijk & Voss, 2010). The time and place of a system's creation process (Kohler et al., 2011; Bitner et al., 2008), customers' needs and processes of service setting (Larsson & Bowen, 1989), and ongoing creation of suitable conditions for value co-generation between a customer and a service provider (Lempinen & Rajala, 2014; Li & Tuunanen, 2022) are key contextual considerations while designing a service system (Seidmann et al., 2011). Customers' positive perceptions can be evoked by attention to details about customers' specific needs and their match with the best possible service alternative (Zomerdijk & Voss, 2010). Creating contextual cues that guide customers' overall service experience is instrumental in driving favorable customer feedback and retention (Mikalef & Krogstie, 2020) through digital transformation of different service elements.

2.1.2. People orientation

People are central to the entire service system experience. Service value co-generation occurs because of interactions between consumers and service providers. Service systems are designed around consumers' needs (Skaggs & Youndt, 2004). Engaging consumers in the development of systems and processed helps create a service offering and delivery mechanism that can cater to their needs (Setia et al., 2013). While consumers are at the center stage of all development service systems, their view can be helpful in designing and digitizing the frontline system for smooth interaction and experiences (Xie et al., 2016; Larsson & Bowen, 1989). In most services, frontline employees/chatbots are the primary drivers of a service system and act as a link between the services offered and consumer satisfaction. This link between services and customer satisfaction offer multiple opportunities for digital transformation not only for the focal organization, but also to diverse stakeholders. Hence, their engagement in the service systems is also significantly important, as this will help in accounting for supply-and demand-side issues (Kohler et al., 2011).

2.1.3. Continual improvements through iterative design

Every system’s digital transformation is iterative (i.e., it involves multiple cycles of feedback and redesign to achieve a design that can match desired functionality expectations). The digital transformation of service systems is also based on continual feedback and improvement cycles based on feedback (Rezwana & Maher, 2022). Incremental improvements in process and service offerings are continually made. This iterative cycle helps incorporate desired changes

according to dynamic consumer requirements. The iterative design philosophy operates by discounting consumer feedback at various stages of service systems (Kautish & Khare, 2022; Peters et al., 2015). Since a service system is an interactive, co-produced offering, it takes into account the vast array of requirements of different types of customers to generate a robust, flexible service design (Lee et al., 2023; Soto Setzke et al., 2023; Setia et al., 2013).

2.2. Cyber-physical systems

Today, generative AI systems complement and support service organizations. Generative AI uses machine learning and natural language processing to absorb and implement computational intelligence that resembles the problem-solving abilities of humans (Chen et al., 2012; Watson, 2017; Kautish & Khare, 2022) and harness the potential of different types of data to come up with insights to solve business and social problems (Ng, 2015). For instance, AI-based chatbots are widely used for interacting with primary customers in different service scenarios such as hospitality, insurance, healthcare, and airlines (Dwivedi et al., 2024; Kautish & Khare, 2022) and are equipped with of machine learning and natural language processing algorithms (Huber et al., 2019; Mikalef & Krogstie, 2020; Tempini, 2017). Another example of harnessing AI and analytics to serve customers better is DBS Bank Singapore, which offers hyper-personalized insights and assists customers in making better financial decisions, ranging from unusual transactions to stock recommendations. This way different technologies are shaping the transformation of servicescape.

Furthermore, since Generative AI is increasingly being used to create customized experiences, service systems must account for functionalities such as real-time personalization and assimilating real-time service-oriented information to create customized service experiences (Chandra et al., 2012; Schreieck et al., 2021). Hence, AI based generative service systems can be defined as "*a service system capable of automating customer service, improving customer experience and enhancing the productivity of customer service team by offering responsive customer service*". Viewing service system concepts through technology-based innovative approaches can transform an entire service experience (Barrett et al., 2015; Lusch & Nambisan, 2015), which can help develop a new classification of CPSC to further advance digital transformation in services. There are evidences in literature that based on specific needs of different stakeholders, factors differ in importance for the adoption of AI systems (Kar and Kushwaha, 2023; Ng, 2015). There is potential for an even more enhanced, holistic system that can handle the complexity arising from variable demand and personalization in services (Soto

Setzke et al., 2023; Lee et al., 2023; Setia et al., 2013). CPS are finding increasing acceptance in the business sphere to drive production, logistics, and supply chain solutions and hold potential for implementation in a service context. Furthermore, CPS are a set of connected systems those are flexible and interactive mode of linking the physical with virtual processes and activities in a seamless fashion. CPS can sense, control, compute, and network with physical infrastructure and objects, linking them to each other through the internet (Versteeg & Bouwman, 2006) and have potential to transform the entire system. These systems utilize the information-processing abilities of a virtual system in conjunction with real-time physical processes and activities to create an inclusive system capable of integrating external knowledge (market research report, business intelligence data, regulatory information, and social media forums etc.) (Chandra et al., 2015). CPS have the potential to incorporate real-time dynamic information by sensing an environment, which, combines historical information, trends and offer solutions for emergent user needs (Nguyen & Nof, 2019).

Moreover, CPS integrate and connect a business's physical and digital processes, resulting in the development of an independent system capable of making decisions and has a high degree of automation (Syed, 2020). Additionally, AI and machine learning induce learning capabilities facilitate ongoing human–machine interactions to become more robust and generate an improved future response (Ng, 2015; Rezwana & Maher, 2022). CPSC have considerable potential for developing novel experiences in logistics, education, healthcare, infrastructure, entertainment, and so on (Brandt et al., 2018; Setia et al., 2011). It can effectively eliminate the need for redundant jobs and substantially improve the effectiveness of service delivery and the overall customer experience (Huber et al., 2022; Lee et al., 2023; Margherita & Braccini, 2020; Setia et al., 2013). Both services and goods industries can reap the potential of these technologies to digitally transform and lead to newer customer experiences (Margherita & Braccini, 2020; Soto Setzke et al., 2023). Data generated from experts' opinions through a grounded theory approach can be helpful in uncovering the social phenomena of elements of a CPSC and how they are organized.

2.3 AI-enabled Generative Cyber-Physical Servicescape

AI-enabled generative CPSC can be termed an advanced version of the service environment that integrates digital and physical domains. With the help of generative AI, CPSC can facilitate creation, adaptation, and optimization, which is adequate. This combination of physical and digital domains offers a more efficient, personalized and engaging experience (Dwivedi et al.,

2024; Ferraro et al., 2024). The AI-enabled generative CPSC has the potential to transform the service sectors such as retail, healthcare, hospitality and transportation (Ooi et al., 2023). Employing AI-enabled generative CPSC in retail can analyze customers' past purchase behavior upon entering the store. Hence it can be helpful to display the personalized offers along with customized shopping atmospheres such as lighting, temperature, and background (Ferraro et al., 2024; Holmström & Carroll, 2024). Further, customers can reduce their time in choosing the clothes by trying them in virtual mirrors as compared to traditional physical settings. In healthcare, AI-enabled generative CPSC can facilitate the creation of personalized treatment plans, and continuous monitoring, eliminating the need for nurses for constant monitoring (Ooi et al., 2023). In the hospitality industry, AI-enabled generative CPSC finds its applications in offering virtual concierge services such as guest inquiries, updating of the housekeeping schedules, monitoring the cameras and sensors for any potential threats and guest safety (Dwivedi et al., 2024). Moreover, AI-enabled generative CPSC has the potential to transform traditional transport to innovative service system by enabling dynamic route adjustment, and integration of different modes such as train, bus, bike and taxi (Richey Jr et al., 2023). In this way, AI-enabled generative CPSC can create a responsive, more innovative and highly personalized service experience that results in offering innovative solutions and enhanced customer satisfaction (Holmström & Carroll, 2024; Ooi et al., 2023).

3. Research design

3.1. Data collection

Since our research focused on an in-depth understanding of the key elements to be considered while designing a AI-enabled generative CPSC for transformation in services, we gathered data via open-ended responses. The data collection included 30 key executives involved in decision-making in service systems development and members of research and development teams. Our semi-structured interview questions revolved around gaining insights into service elements for an organization if it transitions from a servicescape to a CPSC and the scope for generative AI. While developing the interview questions, we ensured objectivity by using natural language and did not use any language to influence the respondents. We have also informed the participants what is the purpose of the research and they have the right to withdraw if they are not willing to participate. Interview responses were collected from organizations providing services in digitized forms, with the potential to employ generative AI in CPSC in different industries. These industries include telemedicine, retail, entertainment, consulting, hospitality, transportation and logistics etc. The respondents profile includes top professionals such as

CEOs/MDs and engineers as presented in Appendix A. These respondents' perceptions help provide direct and firsthand insights critical for real-world impact that technologies are creating in the space of generative AI for CPSC. The qualitative data collected offer comprehensive view to answer the research question. Each interview lasted an average of 50 minutes. The interviews were transcribed, coded, and analyzed using thematic analysis to highlight major themes (Strauss & Corbin, 1990). This approach balances the induction approach and rationality in the early structure (Langley, 1999). Refer to Appendix A for the details of respondents and Appendix B for the semi-structured interview protocol employed. Figure 1 highlight the research design adopted in this study, where four phases highlight the planning, conducting, analyzing and conducting.

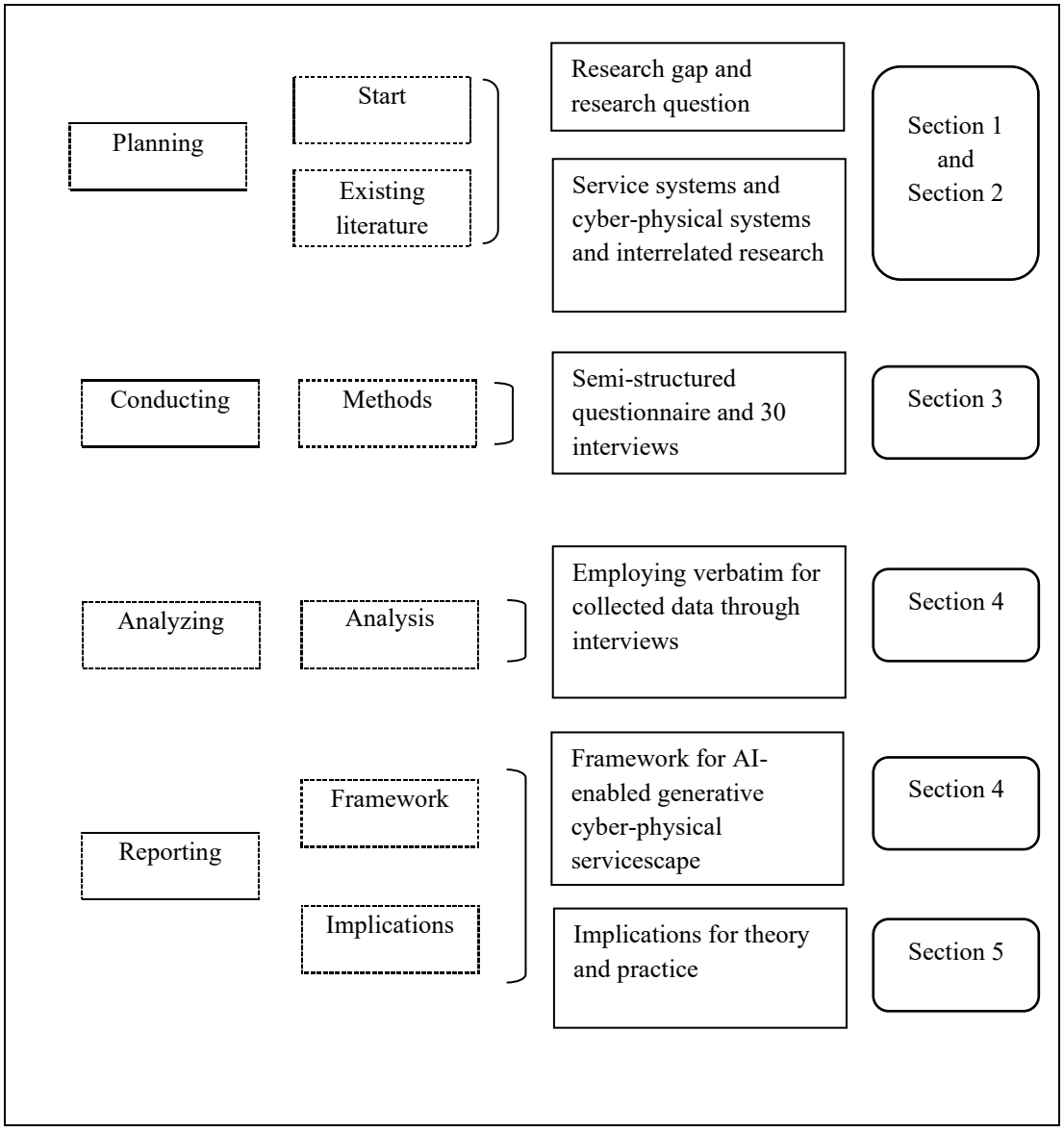


Figure 1. Research Design

3.2 Data analysis

The grounded theory approach is employed to analyze the data to develop concepts and relationships among the data blocks (Glaser & Strauss, 1967). Specifically, this study performed data analysis in line with data collection to take advantage of flexibility in the case research method (Strauss & Corbin, 1998). The Straussian approach to grounded theory was utilized because of its epistemological feasibility to create as compared to discover (Kenny & Fourie, 2015) theory as a starting point of inquiry (Strauss & Corbin, 1990). This approach enables theoretical sensitivity that helps theorize and abstract raw data.

First, we conceptualized and extracted the dimensions and themes of the initial theoretical lens (see Table 1). User interface indicates the point of interaction between a provider and a user, whereas cybercore indicates critical aspects such as storage, sensing capability, and security features. Servicescape refers to the physical environment, where service is being offered, and it includes the service settings that influence the customer experience, and have key features like, spatial layout, ambient conditions, signs, symbols, and artifacts. On the other hand, AI-enabled generative CPSC employ smart technologies in the physical and digital service environment to create, adapt, and optimize interactions through real-time user inputs and data to offer personalized and interactive services. With the progress of primary and secondary data collection, the data were coded in a structured way using techniques such as open, axial, and selective coding (Strauss and Corbin, 1990).

Table 1. Dimensions and themes of the initial theoretical lens

Dimension	Themes
User Interface	User-friendly interaction, designed process, less complex, ability to co-generate
Cybercore	Cloud storage, sensing capability, robust and secure server, physical asset
Servicescape Design	Customer-centric, customer satisfaction, improvement, service environment
Cyber-physical servicescape	Real-time information, smart and autonomous service systems, processing abilities, generative intelligence of integration with external knowledge

First, open coding is used to develop the conceptual understanding of data from the expert interviews, which helped form first-order concepts (Gioia et al., 2013). The first-order concepts

are classified into second-order themes via axial coding. Suppose a first-order concept is fitted to an existing second-order theme within a system; then, the concept is assigned directly. Otherwise, if the fit is not exact or no concept exists for matching, either an existing second-order theme is modified, or a new second-order theme is created (Gioia et al., 2013) before starting the coding again. In the last stage of the grounded theory approach, selective coding is used to classify the second-order themes into aggregate dimensions (Strauss & Corbin, 1990). The axial codes are assessed and re-assed many times to form a related category. For instance, experience offered in a service, co-generation abilities, and effortless navigation opportunities define the user-interface. This analysis establishes the relationship between second-order themes and aggregate dimensions in the coding structure wherever required. Abstracted first-order concepts, second-order themes, and aggregate dimensions (Gioia et al., 2013) are presented in Appendix C as the sample.

A visual mapping approach presents the concepts, themes, and dimensions in one frame derived from the coding structure and theoretical lens (see Figure 2). We used the narrative approach to theorize further the relationships between aggregate dimensions and second-order themes (Langley, 1999). Narrative and visual maps are verified with ten key informants and further refined to validate interpretations and theoretical ideas. We adopted credibility, transferability, confirmability, and dependability criteria (see Table 2) to ensure the rigor of the adopted research methods (Lincoln & Guba, 1985).

Table 2. Measures to ensure research rigor

Criteria	Measures Undertaken
Credibility	Data was collected from a variety of sources, such as industry reports, research articles, and informants (Klein & Myers, 1999) Semi-structured questions were prepared as open-ended and appropriate to the phenomenon under study (Myers & Newman, 2007)
Transferability	The theory that emerged from the data was compared with existing research to facilitate the generalizability (Lee & Baskerville, 2003)
Confirmability	The interview process involved a set of four researchers, and data were analyzed jointly and discussed several times to validate an interpretation (Pan & Tan, 2011)
Dependability	Data collected were compared with primary and secondary sources to ensure accuracy (Klein & Myers, 1999). Emerging themes and constructs were consulted and verified repeatedly to align a theory and the framework (Pan & Tan, 2011)

Data analysis resulted in the generation of insights in multiple directions; this facilitated further data collection from one iteration to another until theoretical saturation was reached (Eisenhardt, 1989). This process supported the development of the framework based on the data collected, themes, and dimensions that emerged (Glaser & Strauss, 1967).

4. Findings

The analysis of the study indicates the emergence of four key themes. The relationship among the key themes proposes a process model for how an AI-enabled generative CPSC emerges from the user experience and digitally transform it. Emergence, the process that indicates that a cumulative framework arises from minor elements (Kenny & Fourie, 2015; Klein & Myers, 1999), is key to complexity science. Following the grounded theory approach, four stages of cyber-physical servicescape are emerging: (i) *user experience,* where customers experience online and offline platforms, generative AI systems and processes co-generate services with organizations and systems; (ii) *cybercore,* where enterprises store and process data generated from initiating to its completion (iii) *servicescape design,* in which context the degree of customer-centricity services are offered; and (iv*) AI enabled-cyber-physical servicescape,* where services and their related elements are offered in real-time and facilitate autonomous decision-making along with a system's high processing abilities.

As presented in the framework, customers' increasing need and expectations trigger the emergence of an AI-enabled generative. It utilizes the principles of CPS to create a user-friendly, robust, customer-centric service system having multi-dimensional processing and sensing capabilities. The section below presents the open, axial, and selective codes of different elements of AI-enabled generative CPSC.

4.1. User interface

The first step in the emergence of CPSC is the user interface, a seamless, creative, and efficient system where customers and providers can interact. User interface and user experience designers work together to develop a service-oriented system. User interface covers appearance, feel, typography, etc. In contrast user experience design focuses on enhancing the satisfaction of customers by refining the usability and accessibility of the services (Wulf & Blohm, 2020), while adopting the digital transformation. Effortless navigation is key to offer seamless experience and creating value collectively through the platform and users (Giesbrecht

et al., 2017). The aggregate dimension, second-order themes (Gioia et al. 2013), emerged in the user interface stage, and their supporting evidence is presented in Table 3.

Table 3. Quotes that emerged as second-order themes in the user interface

Aggregate Dimension	Second-Order Theme	Evidence
User Interface	Experiential	*We operate in the financial service sector, where every day we strive to enhance the experience of our stakeholders, be it employees, customers, brokers, or any other third party involved. Personalization is the new competitive dimension (R7).* *We first understand the user/customer needs, demands, and wants that can further help to develop and design a system and architecture that offers a customer-made experience (R27).*
	Effortless navigation	*When we analyze the design of a platform or system, I particularly look at the touch points that a customer or client has to go through before a service is rendered. Additionally, the platform need to ensure effortless navigation for customers (R3).* *Designing a AI-enabled generative service system need support from Internet of Things (IoT), with a strong association between physical, networking, and computation processes. These processes need to have strong security features and offer immersive visualization (R11).*
	Co-generation	*Today to deliver value to the customer, one needs to involve the customer right from service conceptualization and develop an architecture to support AI-enabled generative services through IoT-like technologies to co-generate both public and private value (R23).* *The customer needs to be involved from the point where he or she looks for a solution rather than a mere service. The generative AI can offer unique solutions for customization and its integration capability makes it above the standalone services offered by other mediums (R5).*

4.2. Cybercore

Along with physical servers, invisible storage solutions like the Cloud are of immense benefit to enterprises to align the services' inadequate flow (Qadeer & Lee, 2023). The processors and

storage mechanism, be it physical or cloud, act as a base that offers a simultaneous quality of service. Adequate storage, memory, and sensing capabilities ensure timeliness, reliability, and security (Syed, 2020).

Cybercore can be considered a core support enabling decision-making for service companies at the front end and customers interacting with service intermediaries. With the increase in the number of data points and data centers worldwide (Mikalef & Krogstie, 2020), demand for a strong foundation of cybercore elements is increasing, as explained by the open quotes in Table 4.

Table 4. Quotes emerged as second-order themes in the cybercore

Aggregate Dimension	Second-Order Theme	Evidence
Cybercore	Server	*To make the CPSC work, there needs to be integration among embedded systems, sensors, actuation, and network systems. Through this integrative system, the cyber-physical system collects data from physical systems for further processing (R19).* *To make it work, the cyber-physical system contains hundreds of sensors to solve complex tasks. To control and coordinate the operations of physical and engineered systems, one must identify the type of sensor or service processes and service logs generated and handled (R7).*
	Sensor	*Today for value-added service, you need a sensing technology that can work on graphics, videos, and market signals along with general-purpose microprocessors. With the humongous data at hand, it is desired that a system be able to execute and sense the business environment from multiple networks (R1).* *Sensors identifying the characteristics of massive and multidimensional data emerging through diverse hardware architectures are required for specific kinds of processing, and how those can be integrated is a challenge for any cyber-physical system. The generic and general-purpose sensors are homogenous and are mostly unable to support complex and heterogeneous cyber-physical system applications (R10).*
	Cloud storage	*Even though we now have a cloud storage facility for the data, many of our partners and stakeholders have their local servers for the storage and access of data.* *Adequate data storage defines efficient management in today's data-driven world (R5).*

	We maintain patients' data for a large hospital chain on our cloud server. This data goes under preprocessing for the area of interest, and further, we store this data on the cloud as well, which is retrieved through a logical algorithm and made available to the clinicians as and when required for further diagnosis (R24).

4.3. Servicescape design

A customer will accept the services if offered in the right environment that appeals to customer pleasure and arousal. This environment further facilitates ease of operation and drives efficiency. Today, services coupled with physical and online platforms make sense and focus on the context in which one offers the services. Customers are continuously changing their expectations; hence, services must continually improve to meet customers' changing expectations (Xie et al., 2016). The customer-centric approach and solutions can be helpful, whether it is data protection, reducing biases, or ensuring an individual's privacy to develop a meaningful servicescape. Table 5 highlights the aggregate dimensions and second-order themes of servicescape design.

Table 5. Quotes emerged as second-order themes in the servicescape design

Aggregate Dimension	Second-Order Theme	Evidence
Servicescape Design	Context	*The service elements are contextual. For instance, offering a service in a retail store and a hospital can have different ambiance and how consumption pattern can be extracted to developed new offering. However, ultimately, the objective is customer satisfaction only in both cases (R27).* *When we want to draw customers' intention to a particular aspect, we focus natural or artificial light on it, whether we are offering physical services or online. In online platforms, it all depends on how you want to navigate the customer through your service portfolio and it can be different for different customers, hence use of generative AI models can be helpful (R9).*
	Customer-centered	*Different services have different designs and landscapes, be they physical or virtual or a combination of both. For instance, one of our customers is in the personal hygiene space, and they have designed their platform to suit each type of customer best. Hence, they first interact online and deliver services physically, and both stages are designed carefully to achieve customer satisfaction through digital transformation (R22).*

		We are a service provider for a retailer who cater to both low-income and high-income customers. They have focused on first exterior design to interior design of service facility, which acts as a deciding factor for most of the customers while consuming the offered services and it can be game changer if we can automate and customize it with generative AI (R14).
	Continual improvement	*The feedback mechanism is critical to understanding if the services offered by an organization match up to its standards. An organization like ours has designed mechanisms that encourage customers to suggest areas of improvement (R4).* *Service quality can be refined and improved when an organization continuously tracks and monitors it. In our organization, we continuously track successes and failures, what worked for us and what did not, and then link them to our improvement process (R25).*

4.4. Cyber-physical servicescape

Services offered to customer's result from many invisible efforts backstage and multiple stakeholders emitting and collaborating on different data types. Therefore, to coordinate this multi-stakeholder effort spread over multiple layers of data, we need a cloud computing infrastructure that can be dynamic and in real-time (Qadeer & Lee, 2023). For mundane and repetitive tasks, companies employ robots or chatbots to transform the processes and improve the system's efficiency digitally (Brandt et al., 2018; Soh & Setia, 2022). Today, customers look for quick, relevant, affordable, and simple solutions to their problems. Therefore, an opportunity exists where systems with processing abilities to integrate external knowledge can be developed to act as plug-and-play to facilitate day-to-day operations in the service industry (Nguyen & Nof, 2019). Table 6 indicates the aggregate dimension and the second-order quotes in a CPSC.

Table 6. Quotes emerged as second-order themes in a cyber-physical servicescape

Aggregate Dimension	Second-Order Theme	Evidence
CPS	Dynamic and real time	*Services today need multi-stakeholder coordination, multidirectional computing, and data analysis, which are either generated from physical resources or networks and sensors. For instance, we at the back end have to ensure that as and when customers enter or leave a hotel, they need to be informed about the services being offered and the facilities covered in the package (R27).*

	We offer services in healthcare where the diagnostic procedure and apparatus are connected to the intelligent system to offer dynamic parameter monitoring that can save the lives of people. Other types of systems are physical and can help virtually when required; the Apple Watch is a publicly available example (R15).
Autonomous & generative	*We have designed many service systems across the spectrum of our clients, and one thing was very common, how we can automate some of the processes. One classic example is feedback, which should be autonomous and self-driven and can be customer specific using generative AI. For instance, AI based systems can analyze audio or text, video for analyzing the data consumption patterns (R29).* *To integrate physical and cyber-based systems, there is a need for a system that controls and owns the technical infrastructure of a network. Different systems can use generative AI to integrate and interconnect to support autonomous and worldwide data exchange (R12).*
Processing abilities	*In services like travel and tourism, many stakeholders are collaborating at the back-end to offer a seamless experience to tourists. Cyber-physical servicescapes are capable of developing mutual trust among collaborators in their decision-making as well as the precision and speed of the decision-making (R3).* *Cyber-physical system-driven services are based on computational capability, architecture, and execution capability. In services, due to large and different types of data, it becomes apparent that a system can sense, organize, think, and analyze it critically to make certain decisions when operationalizing the services (R27).*
External knowledge	*Since services have been majorly labor intensive and most of the data remains in documents, photographs and physical registers. Hence, how this unstructured data can be converted to meaningful knowledge that can help generating unique services and experience. Employing generative AI systems can solve the problem of converting unstructured data to structured data (R24).* *Technology progressed in automating the service processes slowly from tracking customer consumption, refining demand pattern, but how external knowledge on customer daily routine, other habits can be exploited to generate new and innovative services (R5).*

4.5. Proposition and framework development

The emerging cyber-physical nature of service systems demands an effective interface that is user-friendly and auto generates the content, and facilitates services. Developers, providers, and other stakeholders are responsible for developing a comprehensive user interface (Xie et al., 2016). For end users, the Web, mobile applications, blogs, social media, interactive content on the Internet, and chatbot interactions are critical to defining their experience with a business (Forsgren & Byström, 2018). Most organizations use the Internet to inform and co-generate an interactive experience with their offerings (Ooi et al., 2023; Setia et al., 2013). However, now, with chatGPT-like generative AI models, organizations can use dispersed explicit knowledge much more effectively to communicate the offering to enable unique real-time experiences and resolve queries quickly and efficiently (Dwivedi et al., 2024; Dowling & Lucey, 2023).

Customers use platforms, mobile applications, and websites as a medium to seek and find information. Customers, providers, and third parties use these systems to customize and optimize their operations further, leading to customer satisfaction (Rezwana & Maher, 2022). Providers and developers must continually improve back- and front-end processes so that users can engage with them effectively in a way that is parallel to the real world. In the words of R26: *In developing AI-enabled generative CPSC, one has to take care of cognitive (usability and functionality) and dynamic (fun and enjoy) elements to have a robust user interface*. Different industries are developing user interfaces ranging from medical devices and automobiles to hospitality to co-generate unique experiences (Dwivedi et al., 2024; Priharsari & Abedin, 2021). Hence, it is proposed:

P1: ***An AI enabled generative CPSC offer seamless interface to offers unique experience, effortless navigation and co-generation of services.***

System developers must plan for the storage, processing capabilities, and maintenance of servers, sensors, and networks to ensure access to data (Lim et al., 2018). With the rise in data generation and consumption, cloud storage has become critical compared to traditional hardware-based storage (Qadeer & Lee, 2023). Hence, a cloud-based storage can be leveraged by AI-enabled generative CPSC to innovate and offer services seamlessly. Cloud storage helps enhance the scalability, durability, and agility of data storage infrastructure, allowing anytime and anywhere access to big data, which is core part of AI-enabled generative systems (Tempini, 2017; Soh & Setia, 2022; Xie et al., 2016). Intelligent devices like sensors and system sensing capabilities also help classify unstructured data, such as photos, videos, and Web content, that can be further stored in user-friendly formats (Schreieck et al., 2021). Servers are key resources

for ensuring data availability and network data delivery in services. Today, third-party cloud providers are available and offer a pay-as-you-use model, enhancing the flexibility of AI-enabled generative CPSC (Chen et al., 2012; Qadeer & Lee, 2023). In the words of R17: *To conduct the services smoothly in the online environment, the systems must be supported by advanced technologies, such as cloud storage, generative AI systems, chatbots, robots, and robust servers.* The combination of cloud storage, sensors, generative AI systems and robust devices offers enormous scalability and unique characteristics of a CPSC. Hence, it is proposed:

P2: ***Cloud computing infrastructure (Cloud storage, generative systems, sensors, and servers) plays an important role for developing a strong cybercore for a AI- enabled generative CPSC.***

The approach and concept of digital services is influenced by customer choices, platforms, and their degree of integration into the brick-and-mortar model (Xu et al., 2013). It includes ambient conditions, spatial layout, and symbols in physical format. The context and concept of the services being offered differ geographically; hence, the performance of services depends upon the cybercore and user-interface flexibility and their integration into the servicescape (Zomerdijk & Voss, 2010). For example, a consumer can purchase travel tickets and grocery items on the same platform and use the same payment gateway. Hence, it is critical to understand the context as customer-centric as possible and improve continuous storage, analyses, and data management capabilities. The AI- enabled generative CPSC have processing capabilities to offer context-centric services integrating external knowledge such as market research reports, business intelligence data, regulatory information, social media forums, etc. Numerous service systems also include products; hence, these platforms need to integrate third parties and multi-dimensional data from manufacturers and suppliers to offer a smooth experience (Li & Tuunanen, 2022; Margherita & Braccini, 2020). In this way, context familiarity focuses on continuous service improvement, and customer-centric service is critical, where generative AI-enabled generative CPSC can be of much help. Therefore, it is proposed that the following:

P3: ***AI- enabled generative CPSC can facilitate customer- and context-centric continuous improvement of servicescape.***

Seamless delivery of services results from many stakeholders collaborating with developers, cloud storage providers, data management companies, and many more (Schreieck et al., 2021; Tempini, 2017). To provide an authentic and dynamic experience to end customers, chatbots, robots, and generative AI technologies are in place to support, auto-generate and process the information required at every step of service generation and consumption (Dowling & Lucey, 2023; Nguyen & Nof, 2019). AI-enabled generative CPSC is intelligent enough to interpret the input based on the question that supports the decision, solves the problem, and supports the co-generation of the customer experience (Brandt et al., 2018). In the words of R9: *To handle diverse and large customer bases with their unique requirements, the generative systems can provide personalized and anthropomorphic real-time information to both the consumer and the provider, hence the need to develop an AI-enabled generative CPSC.* In large database handling and offering unique experiences to the customer, the processing abilities of generative systems are defined by their networking and computing capabilities. In this way, real-time and dynamic processing, autonomous & generative, and multidirectional processing abilities are key characteristics of a cyber-physical servicescape defining digital transformation. Therefore, it is proposed that the following:

P4: ***AI- enabled generative CPSC facilitate dynamic, real-time, autonomous service generation having multi-directional processing abilities.***

The user interface sets the base for experience and co-generation. Apart from the user interface, an AI- enabled generative CPSC is capable to extract external knowledge (market research report, business intelligence data, regulatory information, and social media forums etc.) to design and deliver a better service experience. It also defines effortless navigation (Lempinen & Rajala, 2022) and the cloud storage and sensing capabilities that influence the context, degree of customer centricity, and continual improvement. Based on the findings, we argue that this process converges into a dynamic, autonomous, and generative system presented in Figure 2.

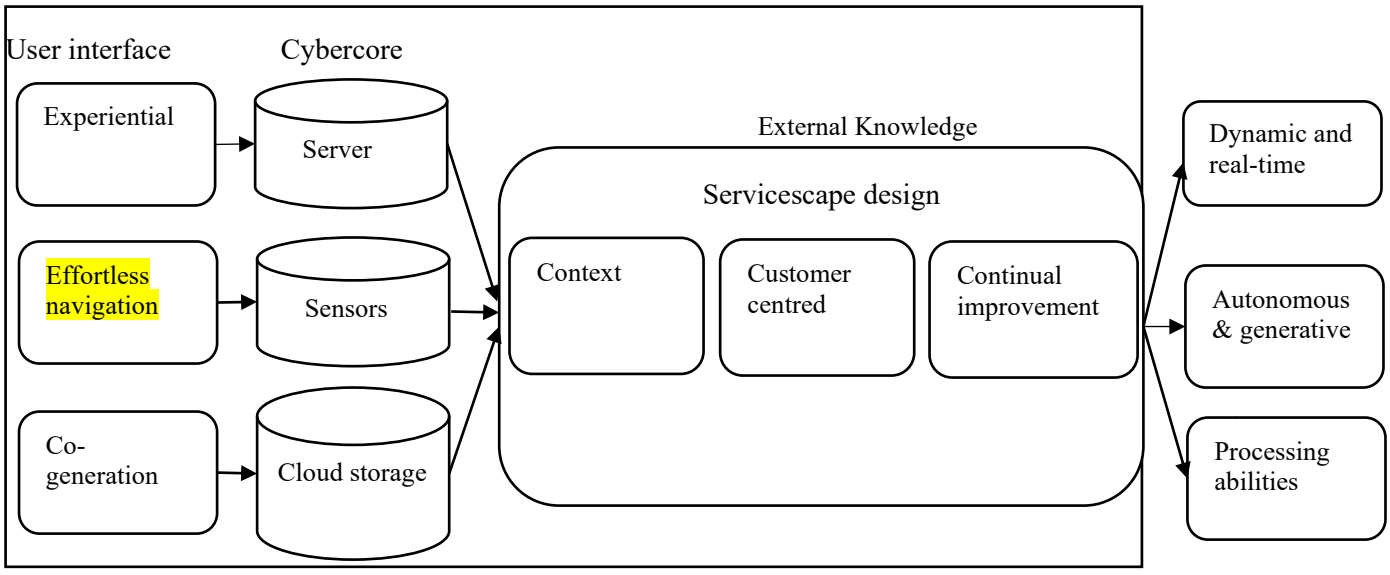


Figure 2. AI-enabled generative CPSC

5. Discussion

The findings highlight the organizational-level elements of service systems and the development of independent and adaptive service experiences. This study also highlights the organization of AI-enabled generative CPSCs. A transition from a traditional service system to AI-enabled generative CPSC entails little or no human contact at the point of service encounter while ensuring collaboration and co-generation, integrating enormous data towards enhanced service experience (Margherita & Braccini, 2020; Tuunanen et al., 2019; Setia et al., 2013).

Our research contributes to understanding potential issues in designing a service system that can foster the autonomous generation and functionality of the servicescape and simultaneously maintain the standards of a personalized service experience by employing AI-enabled generative CPSC (Chuang & Lin, 2017; Dwivedi et al., 2024; Wulf & Blohm, 2020). Service systems are inherently complex, and a lot depends on the design, flow, and process in which the consumer and environment interact and define the consumer experience (Zomerdijk & Voss, 2010). In services, generative AI can be help facilitate mundane tasks such as content generation and quick response and improving service professional's productivity (Dell'Acqua et al., 2023), where other studies have embraced role of generative AI in different sectors from supply chain to hospitality (Dwivedi et al., 2024; Ferraro et al., 2024; Ooi et al., 2023; Richey Jr et al., 2023). Thus, there is a need to develop and generate a flexible and adaptive service system that can work in a dynamic environment (Ostrom et al., 2015). Considering this, the present study contributes to the existing knowledge about technological architecture and service system-related aspects from an organizational view (Versteeg & Bouwman, 2006), which can be useful while devising a transition strategy to an AI- enabled generative CPSC.

5.1. Implications for theory

This study enhances the understanding of service system research, as it is shifting to a new mode of service experience with AI- enabled generative CPSC. The existing literature on service experience and digitalization has considered service creation and delivery, in which encounters are digitalized and performed through smart devices (Barrett et al., 2015; Karwatzki et al., 2022; Wulf & Blohm, 2020). However, conceptualizing, designing, innovating, generating, and offering new services using AI-enabled generative CPSC can be a game changer in the industry. AI-enabled generative CPSC can help generate content automatically and adapt to the interaction, context, and orientation to further its service capabilities. AI-enabled generative CPSC are the systems are capable of offering personalized recommendations such as fashion preferences and choice of a vehicle while booking a car. By integrating CPS with generative AI, companies can contribute to enhance the boundaries of service experience and service innovation. The existing studies provide insights into the co-generation of service experiences in technology-mediated or technology-assisted environments (Li & Tuunanen, 2022; Ramaswamy & Gouillart, 2010). Studies are limited to employing generative AI toward services, and other studies have highlighted the use of AI to detect fraud and risk in businesses (Dwivedi et al., 2024; Kar & Kushwaha, 2021; Ng, 2015).

This study expands the service experiences and service innovation boundaries to include a seamless human/consumer-technology interaction scenario inspected from a service system perspective. Thus, this study can be instrumental in providing exploratory insights into the relevant operational considerations required for a shift to an AI-enabled generative CPSC, where external knowledge (such as market research reports, business intelligence data, regulatory information, and social media forums etc.) can be integrated to create unique services and innovative service experiences. Since design is an inclusive activity that considers all the potentialities and constraints (Najjar et al., 2021), understanding the operational constraints and re-integration requirements can contribute to the conceptualization, design and AI-enabled generative CPSC employment in the industry.

This study contributes to the literature in several ways. *First*, it generates the key themes and subthemes in answering the research question: *What are the key elements of a AI-enabled generative Cyber-Physical Servicescape, and how should they be organized?* Key themes and subthemes include a user interface (experiential, effortless navigation, and co-generation), cybercore (sensors, servers, and Cloud storage), and servicescape design (context, customer centricity, continual improvement) leading to a *AI-enabled generative CPSC* (dynamic and real-time, autonomous, and has processing abilities to integrate external knowledge). *Second*,

based on themes and subthemes, this study drives four propositions that are established through the grounded theory approach. *Third*, following these propositions, this study developed a AI-enabled generative CPSC framework to conceptualize the findings. The framework can be applied to support and enable the design, development, and deployment of innovative service systems. *Finally*, this study will be instrumental in creating an interdisciplinary area of inquiry at the intersection of information systems, organization studies, service research, and design studies and renewing the literature on service systems and information systems.

5.2. Implications for practice

A shift from a servicescape to an AI-enabled generative CPSC can logically imply a transformation into newer technologies and processes. Traditional servicescapes often apply historical, rule-based and data-based predictions (Xie et al., 2016). However, AI-enabled generative CPSC enhances the system's capability using big data, external knowledge from multiple sources, and generative AI for real-time data generation and analysis. Consequently, professionals can aim to provide real-time service customization based on dynamic customer data (e.g., personal purchase preferences) along with real-time data (e.g., a consumer's emotional state, environmental context, and sensory and spatial movements) (Chandra et al., 2012). Additionally, practicing managers must consider integration requirements with other service systems that directly or indirectly affect their functioning (Li & Tuunanen, 2022). The findings of the study logically imply that an AI-enabled generative CPSC would necessitate integration with third-party suppliers, such as payment gateway providers and suppliers and vendors, logistic partners, etc., to arrive at an integration mechanism that allows them to access and authorize real-time transactions with a seamless interface at the consumer's end.

Hence, a shift to an AI-enabled generative CPSC would lead to digital transformation at the more significant organizational level, entailing an overhaul of strategy, market positioning, competition landscape, and capabilities, in addition to managing huge costs associated with restructuring and re-skilling staff. Managers must also organize and align resources, capabilities, and processes using AI-enabled generative CPSC to offer the best service experience. Additionally, customer involvement may be helpful when developing a generative service system and how customers interact, think, and operate within a system.

This way, customer involvement can be ensured from conceptualization to continuously improving and adding autonomous and generative features. Hence, the AI-enabled generative CPSC will be more integrated and seamless. Consequently, organizations and managers must develop organizational processes and create mechanisms for integrating and managing the performance of different functional areas (Setia et al., 2011) by employing AI-enabled generative CPSC.

5.3. Limitations and scope for future research

Servicescape aims to ensure that a customer gets a memorable experience. This study focused on understanding what it takes to develop an AI-enabled generative CPSC and how its elements are organized, where the focus is on customer-centricity and employing principles of CPS. Digitally illiterate customers find AI interactions overwhelming, potentially causing disengagement. Hence, this perspective may be considered by future researchers to explore further. The role of architecture and an emphasis on efficiency, effectiveness, and usefulness can be balanced by customer centricity to advance the design of AI-based generative CPSC further. This study proposed a linear framework of AI-enabled generative CPSC, from the user interface to the cyber core and then generative cyber-physical servicescape. However, these could also be viewed as modular layers that loosely interact with each other, with the user interface at the top and the cybercore at the bottom and vice-versa and analyzing the elements interactions.

Furthermore, the contemporary literature does not support practitioners in considering whether another type of structure or system is needed to serve different markets and segments of their customers. A future study could determine whether an AI-enabled generative CPSC has any influence on customers' expectations. Future studies could empirically test this current study's propositions and further refine the organizational elements of an AI-enabled generative CPSC framework. Qualitative studies should be conducted to further refine the findings, where significant respondents from both the customer and service provider sides could be considered to understand the true potential of an AI-enabled generative CPSC.

Declarations

Ethics approval and consent to participate: This research did not involve human participants and/or animals.

Consent for publication: The author(s) give their explicit consent that this research was not conducted in any organization/ company.

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Appendix A. Respondents Profile

Position	Age	Educational Background	Manufacturing	Logistics and Supply Chain	Consulting	Education/research	Government	IT services/software	Total
Director-IT/Head Digital Transformation/ CTO	31–40	Graduate	1					1	2
	41–50	Post-graduate				1		2	3
	51-60	Ph.D.		1					1
AVP/VP/EVP	41–50	Post-graduate						3	3
Consultant	41–50	Post-graduate	1	1	3		1	3	9
Manager/Sr. Manager	31–40	Graduate	1					2	3
	41–50	Post-graduate			1			1	2
Engineer	20–30	Graduate					1	1	2
	31–40	Graduate	1		1			1	3
	41–50	Post-graduate					1	1	2
Total			4	2	5	1	3	15	30

Appendix B. Semi-structured Interview Schedule

1. What are the various kinds of services that your organization provides? What is the extent of digitalization of the services being offered?
2. What specific challenges do digitalization of services present in the context of your organization?
3. Do you think service-based organizations can benefit if they transition to a cyber-physical servicescape from a conventional servicescape? How?
4. What kinds of operational challenges do you anticipate if your organization adopts a more cyber-physical interface for the service process?
5. What are the possible technical issues in creating a flexible cyber-physical servicescape that can adapt to user requirements in varied service contexts?
6. What aspects of your current operational (process) routines might be altered to create a flexible cyber-physical servicescape that can adapt according to users' requirements in varied service contexts?
7. With respect to your organization, what aspects of current technological capabilities may get altered/augmented to create an interactive, independent cyber-physical servicescape?
8. What aspects of back-end operational routines may require restructuring in the case of a shift to a cyber-physical servicescape based on artificial intelligence (AI) learning capabilities?
9. Which technological changes may be necessary in the case of a shift to a cyber-physical servicescape based on AI learning capabilities?

Appendix C. Sample Data Structure

